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Experimental verification and 4E analysis of a hydrate-based CO₂ capture system for flue gas decarbonization

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Abstract

Gas hydrates, crystalline compounds composed of water and guest molecules, have gained attention for their potential in selective CO₂ capture and storage. This study evaluates hydrate-based CO₂ capture technologies for flue gas decarbonization through experimental investigations and process simulations. Hydrate formation and dissociation experiments examined two configurations: a two-stage high-pressure formation process using the kinetic promoter sodium p-styrenesulfonate, which achieved a 53.65% CO₂ removal rate and reduced concentration from 20 mol% to 9.27 mol%, and a three-stage low-pressure formation process employing both kinetic and thermodynamic promoters (TBAB and cyclopentane), attaining a 64.66% removal rate and lowering CO₂ concentration to 9.11 mol%. Complementary to the experimental data, process simulations were conducted by the Aspen HYSYS and Aspen EDR. Then the comprehensive 4E (Energy, Exergy, Economy, and Environment) analysis identified the Low-Pressure Formation with Atmospheric Dissociation (L-A) configuration as the most effective approach. The L-A process exhibited the lowest total energy consumption of 240,077 MJ/h and the highest exergy efficiency of 0.725. Economically, it presented significantly lower equipment and operational costs compared to high-pressure alternatives. Environmentally, the L-A configuration maintained indirect CO₂ emission ratios below one, indicating a net positive impact. These results suggest that the L-A process offers a balanced and efficient solution for industrial-scale CO₂ capture, combining technical feasibility, cost-effectiveness, and environmental sustainability.

Highlights

1. Two- or three-stage hydrate-based CO₂ capture process cut CO₂ from 20% to ~9%.
2. The 4E analysis (energy, exergy, economy, and environment) were conducted.
3. L-A process for HBCC yields lowest energy and highest exergy.
4. A scalable and cost-effective solution for industrial CO₂ capture was proposed.

Keywords Gas hydrates, CO₂ capture, Decarbonation, Process simulation, 4E analysis

1 Introduction

Gas hydrates, crystalline solid compounds composed of water and guest molecules such as natural gas components, have garnered significant attention in both scientific research and industrial applications over the past few decades [16, 51]. Structurally, gas hydrates form under specific conditions of low temperature and high pressure, where water molecules create a lattice that traps gas

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molecules within its cavities [49]. This unique structure not only imparts remarkable stability under the right conditions [7, 14, 17, 47, 54, 57] but also facilitates selective gas capture and storage [24, 61, 67], making hydrates a promising medium for various industrial processes.

The versatility of gas hydrates extends to a broad spectrum of industrial applications. In the energy sector, hydrates are explored for efficient natural gas storage and transportation [59, 63], potentially offering a higher energy density compared to conventional methods. Additionally, gas hydrates play a crucial role in carbon capture and sequestration (CCS) [29, 64]. By selectively trapping carbon dioxide (CO₂) from industrial emissions, hydrates provide a means to mitigate greenhouse gas emissions [13, 46, 72].

Among these applications, the separation of gases using hydrate-based methods, especially for flue gas treatment, has emerged as a particularly promising area of research [61]. Flue gas, primarily consisting of nitrogen and carbon dioxide, and trace amounts of other gases, poses significant environmental challenges due to its contribution to greenhouse gas emissions [20]. Traditional methods for flue gas treatment, such as absorption [19, 36], and membrane separation [27, 43], often suffer from limitations related to energy consumption, cost, and selectivity. Absorption technology [66] is well-established but suffers from high desorption costs, large equipment requirements, low cycling efficiency, and environmental concerns due to solvent pollution. Pressure swing adsorption (PSA) [62] is also a mature technology, though it is characterized by frequent adsorption and desorption cycles, high automation requirements, and the susceptibility of adsorbents to contamination. Adsorptive distillation [25], while effective for producing high-purity products, requires regular replacement of adsorbents and can be energy-intensive. Low-temperature distillation [10], which is well-suited for producing food-grade products, is hindered by large-scale equipment and high energy consumption due to low-temperature operations. Membrane separation [23] allows for multi-stage separation without the need for chemical agents, offering low energy consumption and small footprint; however, membrane materials have a short lifespan and limited durability. In contrast, hydrate-based separation offers a potentially more efficient and selective approach for CO₂ capture, leveraging the natural affinity of hydrates for CO₂ molecules [5, 6, 12, 15, 39].

However, the practical implementation of hydrate-based flue gas separation on an industrial scale faces several challenges. A primary obstacle is the inherently slow rate of hydrate formation, which can impede the feasibility of large-scale applications [22, 53]. To address this, researchers have experimented with various methods

to enhance hydrate kinetics. Mechanical agitation [68, 71] and spraying [31] techniques aim to increase mass transfer rates and promote uniform hydrate formation. Additionally, the incorporation of chemical promoters—such as amino acids [9, 28, 34, 35, 65], tetra-*n*-butylammonium bromide (TBAB) [21, 37], and tetrahydrofuran (THF) [44, 52, 70]—has shown promise in lowering the induction time and increasing the hydrate formation rate.

Despite these efforts, there remains a paucity of comprehensive studies that examine hydrate-based decarbonization processes at an industrial process scale. Most existing research [2, 30, 45, 55] tends to focus on laboratory-scale experiments or theoretical simulations, often lacking a holistic analysis that integrates experimental data with process design and economic evaluation. This gap underscores the need for studies that not only explore the fundamental aspects of hydrate formation but also evaluate the practicality and scalability of hydrate-based gas separation technologies in real-world industrial settings.

To advance hydrate-based flue gas separation technologies, initial high- and low-pressure experiments were conducted to gather essential experimental data. Utilizing Aspen Plus software, four distinct process configurations—High-Pressure Formation with Pressurized Dissociation (H-P), High-Pressure Formation with Atmospheric Dissociation (H-A), Low-Pressure Formation with Pressurized Dissociation (L-P), and Low-Pressure Formation with Atmospheric Dissociation (L-A)—were subsequently developed. A comprehensive 4E analysis, encompassing Energy, Exergy, Economy, and Environment [32, 38, 40, 48], was employed to evaluate the performance and sustainability of these configurations. This thorough assessment identified the low-pressure formation with atmospheric dissociation (L-A) configuration as the most effective for flue gas decarbonization, achieving an optimal balance between technical feasibility, cost efficiency, and environmental impact. The findings enhance the understanding of hydrate-based gas separation technologies and provide practical insights for their industrial implementation, thereby contributing to the development of sustainable and efficient methods for mitigating industrial CO₂ emissions.

2 Hydrate-based CO₂ capture system

This section first introduces the materials and apparatus used in the experiments, followed by a clear explanation of the operational procedures and data processing methods for hydrate formation. Building on this foundation, a conceptual process flow design is proposed, and a detailed model along with corresponding parameter settings is developed with the aid of process simulation tools. Additionally, the calculation principles and

evaluation criteria for the 4E analysis are determined. These preparations establish a set of analytical and evaluative standards that will be applied in the subsequent analysis of hydrate experimental data in Sect. 3, as well as in the discussion of the process simulation results in Sect. 4.2.

2.1 Experimental section

2.1.1 Materials and apparatus

In the hydrate experiments, sodium p-styrenesulfonate (SS, purity > 90%) was employed as the kinetic promoter, while tetrabutylammonium bromide (TBAB, AR, > 99%) and cyclopentane (CP, > 96%) were used as thermodynamic promoters [8]. All these chemicals were purchased from Aladdin. The gas mixture (20% CO₂ and 80% N₂) was supplied by Heli Gas Co., Ltd. In addition, the experiments utilized in-house produced deionized water with a resistivity greater than 18.2 MΩ·cm at 25 °C, ensuring the high purity required for accurate and reproducible hydrate formation.

The experimental setup shown in Fig. 1 consists of a vertical 316L stainless steel reactor (volume: 121.4 mL, bottom diameter: 4 cm, height: 7.8 cm) placed within a temperature-controlled bath (TMS8028-R15), which maintains stable conditions (temperature fluctuation of 0.01–0.02°C). The inlet gas mixture, selected based on typical industrial flue gas compositions and literature surveys [60], contains 20% CO₂ and 80% N₂. A gas chromatograph (GC, SP-3500) is used to analyze the compositions of both the inlet and outlet gas streams, ensuring accurate identification of CO₂ concentration before and after hydrate formation. Temperature and pressure inside the reactor are monitored by a PT100 sensor and

a Trafag 8252.85.2517 pressure sensor, respectively, with data recorded and processed via a Keysight 34970A data acquisition system. This integrated setup provides precise and reliable conditions for investigating the hydrate-based CO₂ capture process.

2.1.2 Procedure and data processing

Before each experiment, the reactor was thoroughly rinsed with high-purity deionized water and then charged with the prepared aqueous solution containing the promoters (SS 8 mmol/L, TBAB 0.29 mol% and 5 vol% CP). The reactor was cooled to 2 °C and subsequently pressurized to either the higher-pressure condition (8.5 MPa) for the kinetic-only promoter scenario or the lower-pressure condition (1.5 MPa) for the combined kinetic and thermodynamic promoters' scenario, with equilibrium pressures determined by CSMGem [1, 16] calculations in Fig. 2. Initially, a 20% CO₂/80% N₂ gas mixture was introduced, but in subsequent cycles the inlet composition was adjusted according to the tail gas from the previous run, thereby emulating industrial operating conditions. For the next experiment, a gas mixture with the specific composition, along with the prepared solution, will be used. Throughout each run, temperature and pressure were continuously monitored, and the gas compositions were analyzed by GC to verify CO₂ capture performance.

To evaluate the effectiveness of CO₂ capture within experimental setup, two primary indicators, reaction off-gas CO₂ concentration and CO₂ capture rate θ , were employed. As described in Sect. 2.2.1, the gas-phase CO₂ concentration can be measured directly using a gas chromatograph (GC). The CO₂ capture rate θ quantifies how

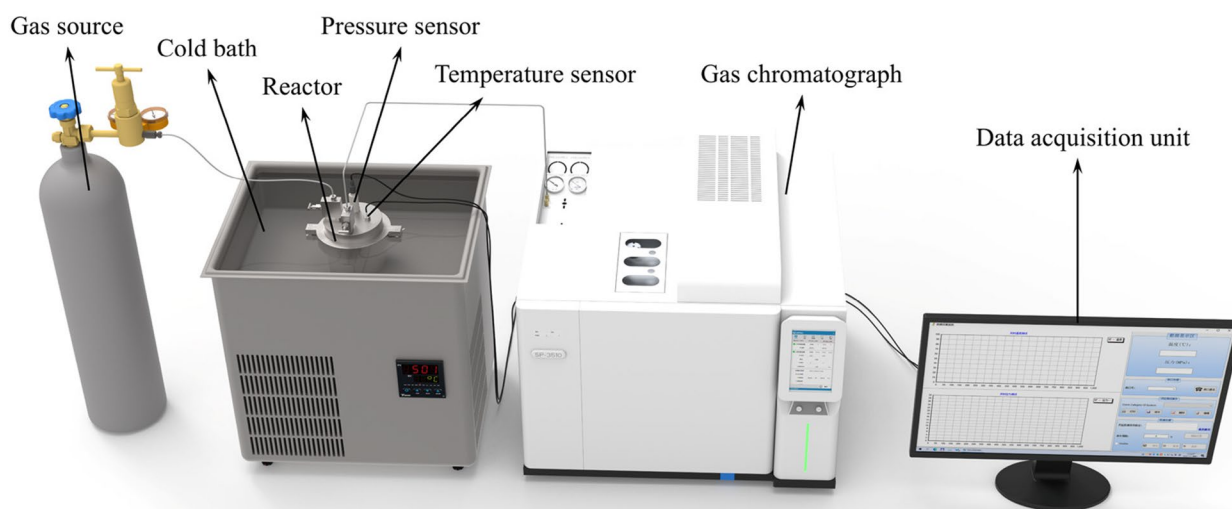


Fig. 1 The schematic diagram of the experimental setup

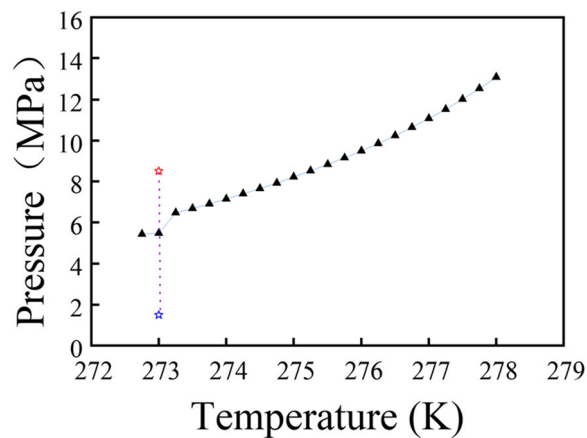


Fig. 2 The hydrate phase equilibrium diagram for an 80% N₂ and 20% CO₂ gas mixture

much CO₂ has been removed (captured) from the original gas feed. Mathematically, it is expressed as:

$$\theta = 1 - \frac{N_{gas,t} * C_{CO_2,t}}{N_{gas,0} * C_{CO_2,0}} = 1 - \frac{\frac{P_0 V_{gas}}{Z(P_0, T_0) R T_0} * C_{CO_2,t}}{\frac{P_t V_{gas}}{Z(P_t, T_t) R T_t} * C_{CO_2,0}} \quad (1)$$

where, $N_{gas,0}$ is the initial total moles of the feed gas, $C_{CO_2,0}$ is the initial CO₂ concentration in the feed gas, $N_{gas,t}$ is the total moles of the off-gas in the reactor headspace at time t , $C_{CO_2,t}$ is the CO₂ concentration in the off-gas at time t , P and T represents the pressure and temperature at the corresponding time, V_{gas} is the gas volume in the reactor headspace, and the compressibility factor was determined via Aspen HYSYS for the gas mixture at specific pressure and temperature conditions.

2.2 Process development and simulation

2.2.1 Conceptual process design

Based on the experimental results in Sect. 2.1, which include CO₂ removal efficiency after each cycle and the composition of the resulting tail gas, a conceptual process design was developed for hydrate-based flue gas decarbonization. Two distinct process configurations

were evaluated. The first process, which utilizes only a kinetic promoter (SS), does not occupy hydrate cavities, allowing for higher single-pass CO₂ removal. This configuration enables the system to meet decarbonization targets with fewer stages; however, it requires higher operating pressures, resulting in greater equipment costs and higher energy consumption. To mitigate this, the second approach combines both kinetic and thermodynamic promoters (SS, TBAB and CP), which reduces the pressure needed for hydrate formation. While this combination reduces the equipment requirements, it leads to a decrease in single-pass CO₂ removal efficiency, ultimately lengthening the overall process.

In addition to hydrate formation, hydrate dissociation is an integral step in the decarbonization process. As indicated by Fig. 2 and basic hydrate equilibrium principles, dissociation at atmospheric pressure requires less thermal input but necessitates re-pressurization of the resulting liquid before it can enter the next hydrate formation stage, thereby increasing total energy consumption. By contrast, performing hydrate dissociation at a pressure closer to the downstream reactor conditions can reduce the pumping work needed to re-establish the desired pressure. This trade-off between dissociation pressure and the associated energy demands must be carefully evaluated when designing a practical and efficient hydrate-based CO₂ capture system.

Based on the preceding analysis, four distinct process configurations emerge: one featuring high-pressure formation with pressurized dissociation, another using high-pressure formation with atmospheric dissociation, a third employing low-pressure formation with pressurized dissociation, and a fourth utilizing low-pressure formation with atmospheric dissociation. For improved readability, these four processes are coded according to their formation and dissociation conditions. The letters “H” and “L” represent high- and low-pressure formation, respectively, while “P” and “A” denote pressurized and atmospheric dissociation. Thus, as shown in Fig. 3, we refer to the four processes as H-P (high-pressure formation, pressurized dissociation), H-A (high-pressure formation, atmospheric dissociation), L-P (low-pressure

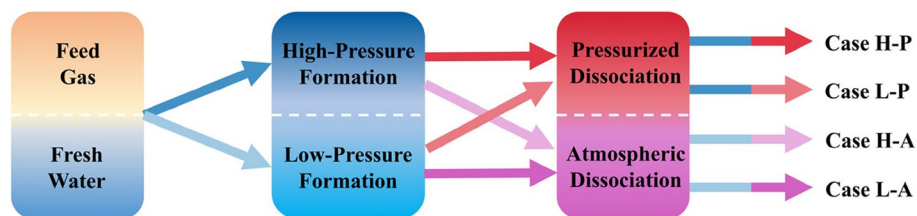


Fig. 3 The schematic diagram of four process flows based on different hydrate formation (high-pressure H and low-pressure L) and dissociation conditions (pressurized dissociation P and atmospheric pressure A)

formation, pressurized dissociation), and L-A (low-pressure formation, atmospheric dissociation).

2.2.2 Process modeling and simulation setup

To establish the hydrate-based carbon capture process, the simulations in this study were conducted using Aspen HYSYS V12. The main challenge in the modeling lies in developing reactor models for hydrate formation and dissociation, while other unit operations such as compression, cooling, and heating [3, 4] can be represented by standard built-in modules. Since hydrates are not available as predefined species in Aspen HYSYS property packages, they must be introduced as hypothetical components, and corresponding formation and dissociation reactions must be defined. Given that the focus here is primarily on material and energy balances, both the hydrate formation and dissociation reactors were modeled as conversion-type reactors, within which the formation and dissociation reactions were specified. As shown in SI Table 1, the key parameters for the hypothetical CO₂ hydrate component were determined based on previous studies and established modeling approaches [11].

2.2.3 Calculation of 4E analysis

To evaluate the effectiveness and practicality of the hydrate-based CO₂ capture process established in the previous sections, a comprehensive assessment beyond material and energy balances is required. Here, the 4E analysis—focusing on Energy, Exergy, Economic, and Environmental aspects [33, 40],—was adopted to provide a holistic perspective on the process performance. The 4E methodology offers a systematic way to identify areas for improvement, gauge economic feasibility, and understand the environmental implications. By integrating these four dimensions, it is possible to balance technical efficiency with economic and ecological considerations, ultimately guiding process optimization and informed decision-making.

(1) Energy analysis

Energy analysis involves quantifying all forms of energy consumed throughout the process. Key steps include condenser heat load, pump power consumption, heater energy input, compressor energy consumption and total energy consumption. All calculation results are derived from the energy flow data in Aspen HYSYS.

(2) Exergy analysis

Exergy analysis is essential for assessing the thermodynamic performance and efficiency of industrial processes

by evaluating not only the quantity but also the quality of energy. Unlike energy analysis, exergy analysis identifies where and how energy degradations occur, revealing inefficiencies and opportunities for optimization. This leads to more sustainable and cost-effective operations. In this study, exergy calculations based on fundamental thermodynamic principles are used to evaluate work interactions and energy losses throughout the decarbonization process. The following equations detail the methodology for quantifying exergy within the system.

The following equation calculates the specific exergy of a system by considering both the enthalpy and entropy changes relative to the environment. It quantifies the maximum useful work obtainable as the system reaches equilibrium with its surroundings [26].

$$Ex_{phys} = -(H_0 - H_1) + T_0(S_0 - S_1) \quad (2)$$

where, H_0 and H_1 are enthalpy of the system at state point of interest and the reference state, respectively; S_0 and S_1 are entropy of the system at state point of interest and the reference state, respectively; T_0 is the absolute temperature of the environment (reference temperature).

To evaluate the exergy efficiency of each equipment within the hydrate-based flue gas decarbonization process, two fundamental equations are employed: the exergy balance equation and the exergy efficiency formula. These equations provide a quantitative assessment of how effectively each device converts input exergy into useful work while accounting for exergy losses due to irreversibility.

The exergy balance equation is essential for determining the exergy interactions within a system or equipment. It is expressed as [26]:

$$Ex_{in} - (Ex_{out} + I_r) = W \quad (3)$$

where, Ex_{in} and Ex_{out} are exergy input and output to the system, respectively; I_r is Exergy loss due to process irreversibility. W is network output of the system or equipment.

Exergy efficiency (η_{ex}) quantifies the effectiveness of a system or equipment in converting input exergy into useful work. It is calculated using the following formula [18]:

$$\eta_{ex} = W/Ex_{total} = 1 - \eta_{ex,loss} = 1 - I_r/Ex_{total} \quad (4)$$

where, Ex_{total} is the total exergy into the system.

(3) Economic analysis

In industrial economic analyses, particularly for pressure vessels such as reactor, the equipment investment cost is significantly influenced by the reactor's shell thickness, its total mass, and the unit price of the construction

material. Below is an overview of the key relationships and formulas used to calculate the minimum required wall thickness, the total mass of the tower, and, ultimately, the total investment cost of the tower.

To ensure the reaction tower safely withstands internal pressure, minimum required thickness was calculated using the following formula [56]:

$$t = \frac{D_i \times P_i}{2S - P_i} \quad (5)$$

where, D_i is the internal diameter of the tower. P_i is the internal operating pressure; S is the maximum allowable stress, typically taken as 1.1 times the operating pressure to account for a safety margin; t is the required minimum wall thickness.

Once the required thickness t is determined, the mass of the tower Q_w can be estimated using [56]:

$$Q_w = \frac{\rho L \pi [(D_i + t)^2 - D_i^2]}{4} \quad (6)$$

where, ρ is the density of the reactor, e.g., 7.85 g/cm³ for steel, L is the height of the reactor; $(D_i + t)^2 - D_i^2$ reflects the cross-sectional area difference between the outer and inner surfaces of the cylindrical shell.

The total cost E of the reactor is composed primarily of the raw material cost and the processing (fabrication) cost, which is often approximately equal to the raw material cost (a 1:1 ratio). Hence, the cost can be simplified as follows [56]:

$$E = (\epsilon \times Q_w) \times 2 \quad (7)$$

where, ϵ is the unit price of the material per unit mass; Q_w is the mass of the reactor.

A commonly used empirical formula for compressor cost is given by [69]

$$C_{compressor} = 580000 + 20000W^{0.6} \quad (8)$$

where, fixed cost component (580,000) represents the base cost of purchasing and installing a compressor system, accounting for standard design, infrastructure, and installation aspects. W is the compressor power. Variable cost component scales with the power requirement of the compressor. The exponent 0.6 reflects economies of scale—as capacity increases, cost goes up but at a rate lower than linear.

For pump investment cost C_{pump} , another empirical formula [58] is

$$C_{pump} = 5040HP^{0.69} \quad (9)$$

where, HP is the pump's horsepower rating. Similar to the compressor equation, the exponent 0.69 indicates

that increasing pump capacity (reflected by horsepower) does not cause costs to rise on a strictly 1:1 basis. Instead, costs grow more slowly due to scale effects in manufacturing and design.

(4) Environmental analysis

Below is the environmental (carbon) analysis portion of a process evaluation. The objective here is to quantify the indirect carbon emissions arising from energy (utility) consumption—particularly electricity.

Indirect carbon emissions C^i can be calculated by summing up the contributions from all utilities and all relevant greenhouse gases (g) [50]:

$$C^i = \sum_k \sum_g C_{k,g}^i \times GWP_g \quad (10)$$

where, C^i represents the indirect carbon emissions due to utility consumption (in kilograms, kg); $C_{k,g}^i$ is the emission factor capturing how much of gas g is produced by utility k . GWP_g is the global warming potential of gas g .

In this work, the only form of energy used by the process is electric power. Consequently, the primary greenhouse gas of concern is carbon dioxide generated by power generation. Therefore, we simplify the equation [41, 42] to:

$$C^i = C_{D,CO_2}^i \times GWP_{CO_2} \quad (11)$$

$$C_{D,CO_2}^i = \vartheta \times E \quad (12)$$

where, ϑ is emission factor for electricity, based on a U.S. standard or relevant regional data. E is the electricity consumption.

3 Experimental results and discussion

3.1 Hydrate formation with only kinetic promoter under higher pressure

The experiment initially investigated the decarbonization performance of the hydrate-based method for removing carbon dioxide from flue gas in the presence of the kinetic promoter sodium styrene sulfonate. The decarbonization results are presented in Fig. 4.

Figure 4 illustrates the CO₂ capture rate and the concentration of CO₂ in the treated gas for the two-stage hydrate decarbonization experiments. Each experiment was conducted six times to ensure accuracy and reproducibility. As shown in Fig. 4a, the first stage of the hydrate experiment achieved a CO₂ removal rate of 39.35 ± 1.21%. Correspondingly, Fig. 4b demonstrates that the CO₂ concentration decreased from the original

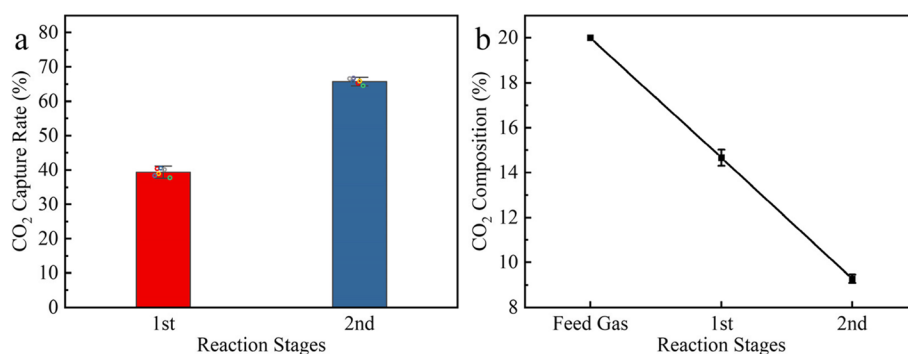


Fig. 4 The decarbonization effect of the hydrate-based method. **a.** The CO₂ capture rate after the first and second stage of hydrate formation. **b.** The CO₂ concentration in the gas phase of the feed gas, after the first and second stage of hydrate formation. (T: 273.15 K, P: 8.5 MPa)

20 mol% in the feed gas to 14.66 ± 0.35 mol%, representing a reduction of 36.7%.

To simulate real industrial application scenarios, the gas composition for the second hydrate reaction was based on the tail gas composition from the first reaction. Further analysis in Fig. 4. reveals that after two hydrate reactions, the CO₂ removal rate reached $65.8 \pm 0.83\%$. Additionally, the final tail gas CO₂ concentration was further reduced to $9.27 \pm 0.18\%$, achieving an overall decrease of 53.65% compared to the initial feed gas.

3.2 Hydrate formation with kinetic and thermodynamic promoter under lower pressure

The experiment further investigated the hydrate-based decarbonization process for flue gas by introducing both the kinetic promoter sodium styrene sulfonate and a thermodynamic promoter (TBAB and CP). To achieve a decarbonization effect comparable to that depicted in Fig. 4, a three-stage hydrate reaction process shown in Fig. 5 was necessary. In the first stage, the CO₂ removal rate was $32.51 \pm 0.46\%$, reducing the CO₂ concentration in the treated gas from the initial

20 mol% to 14.94 ± 0.33 mol%. In the second stage, the removal rate increased to $50.55 \pm 0.41\%$, further decreasing the CO₂ concentration to 11.68 ± 0.28 mol%. Finally, the third stage achieved a CO₂ removal rate of $64.66 \pm 0.35\%$, lowering the tail gas CO₂ concentration to 9.11 ± 0.37 mol%.

When comparing the three-stage process in Fig. 5 to the two-stage system without the thermodynamic promoter as shown in Fig. 4, several key observations emerge. The introduction of the thermodynamic promoter necessitated an additional reaction stage due to its incorporation into the hydrate cages, which inhibited the efficiency of CO₂ capture in each individual stage. The first stage with both promoters exhibited a lower CO₂ removal rate of $32.51 \pm 0.46\%$ compared to $39.35 \pm 1.21\%$ without the thermodynamic promoter. However, by extending the process to three stages, the overall CO₂ removal rate increased to $64.66 \pm 0.35\%$, and the final tail gas CO₂ concentration was reduced to 9.11 ± 0.37 mol%. This performance is comparable to the two-stage system without the thermodynamic promoter, which achieved a

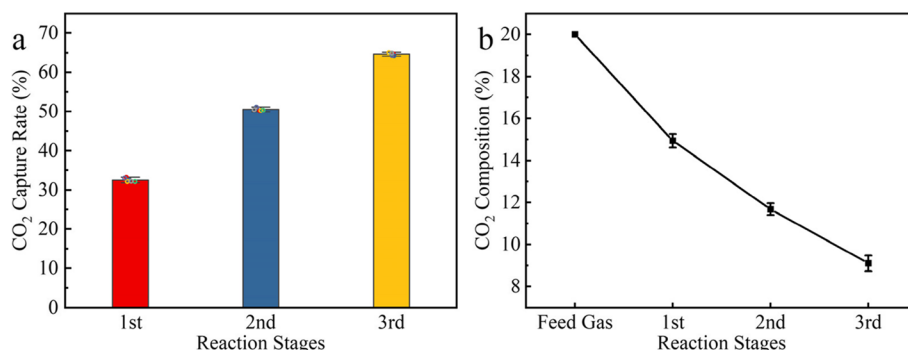


Fig. 5 The decarbonization effect of the hydrate-based method with thermodynamic promoter. **a.** The CO₂ capture rate after the first, second and third stage of hydrate formation. **b.** The CO₂ concentration in the gas phase of the feed gas, after the first, second and third stage of hydrate formation. (T: 273.15 K, P: 1.5 MPa)

$65.8 \pm 0.83\%$ removal rate and a final CO_2 concentration of $9.27 \pm 0.18\%$.

4 Results and discussion of process simulation

Based on the experimental results presented in Figs. 4 and 5 of Sect. 3, it was determined that achieving the desired level of decarbonization using the hydrate-based flue gas method depends significantly on the operating pressure. Under higher pressure conditions, two reaction stages are necessary to meet decarbonization requirements. However, when operating at lower pressures, the conversion rate in a single reaction stage diminishes, necessitating the implementation of three reaction stages to achieve comparable decarbonization performance. This scenario presents a fundamental balance between high-pressure and low-pressure processes. High-pressure operations, while enabling shorter process flows, entail increased manufacturing and operational costs due to the need for robust single-stage equipment capable of withstanding elevated pressures. In contrast, low-pressure processes involve longer process flows and additional reaction stages, but they benefit from lower equipment and operational expenses, presenting a trade-off between capital and operational expenditures.

In addition to the reaction stages, the dissociation process plays a critical role in the overall system efficiency. As illustrated in Fig. 3, two dissociation strategies are evaluated: constant pressurized dissociation and atmospheric pressure dissociation. Constant pressurized dissociation is characterized by lower electrical energy

consumption for pumping, as maintaining a steady pressure reduces the energy required for compression. However, this approach demands more extensive cooling and heating utility to manage the thermal loads associated with maintaining constant pressure. Conversely, atmospheric pressure dissociation minimizes the reliance on cooling and heating utilities, thereby reducing associated costs. Nevertheless, this method results in higher energy consumption for pumps, offsetting some of the utility savings.

To comprehensively evaluate the four proposed process configurations, Sect. 4.1 utilizes Aspen HYSYS V10 to conduct full-process simulations for each scenario. This simulation framework allows for detailed modeling of the hydrate-based decarbonization process under varying operational conditions. Following the simulation studies, Sect. 4.2 performs an in-depth 4E analysis—covering energy, exergy, economic, and environmental dimensions—for all four process configurations. This multifaceted analysis provides a holistic assessment of each process's performance, identifying key areas of efficiency and potential cost savings while also evaluating environmental impacts.

4.1 Description of Hydrate-based CO_2 capture process

4.1.1 The process of high-pressure hydrate formation and pressured dissociation (H-P)

The process flow depicted in Fig. 6 is divided into two primary sections: the gas and liquid cooling and

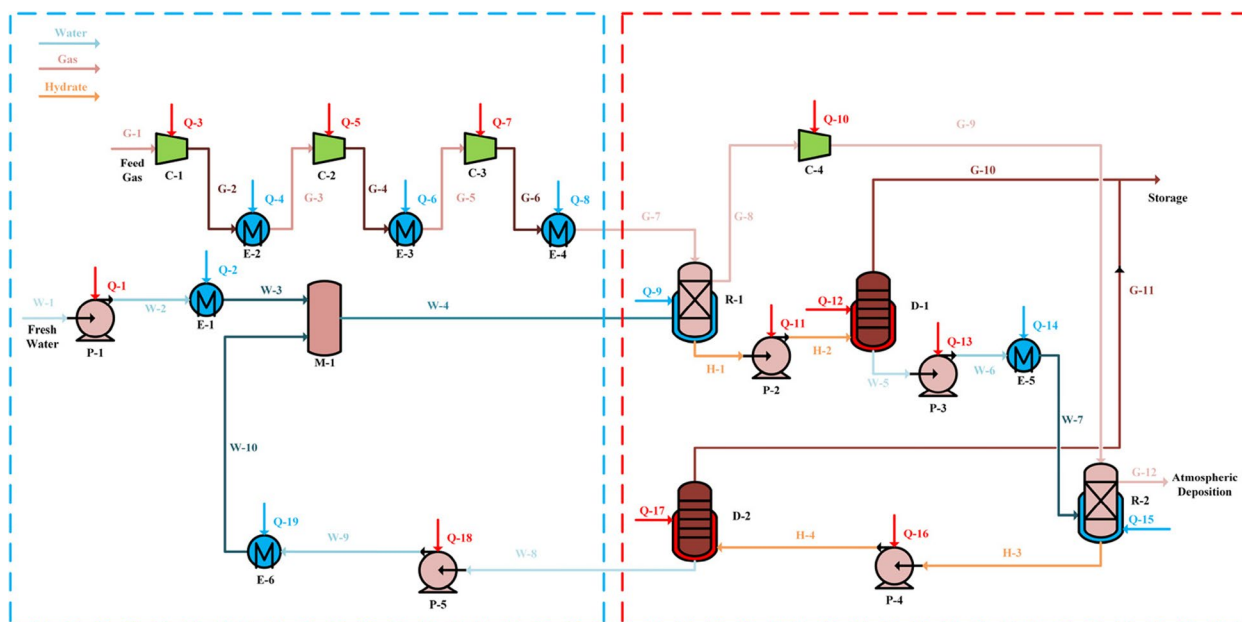


Fig. 6 The process flow diagram of high-pressure hydrate formation and pressured dissociation

pressurization section on the left, and the hydrate formation and dissociation section on the right.

In the left portion of the process flow, the raw gas (G-1) undergoes a three-stage compression process facilitated by compressors C-1, C-2, and C-3. Following compression, the gas is cooled through a series of heat exchangers (E-2, E-3, and E-4) to achieve the desired conditions of 8.5 MPa and 0 °C. Simultaneously, fresh water (W-1) and recirculating water (W-8) are each subjected to pressurization and cooling. These treated water streams are then combined to form the high-pressure, low-temperature reaction liquid (W-4), which serves as the medium for hydrate formation.

The right side of the process flow involves two stages of hydrate formation and dissociation, designated as the first stage (R-1 and D-1) and the second stage (R-2 and D-2). A key feature of this design is that both the hydrate formation reactors (R-1 and R-2) and the dissociation reactors (D-1 and D-2) operate at a consistent pressure of 8 MPa.

Due to the requirements of pressurized dissociation, higher decomposition temperatures are necessary. Consequently, both D-1 and D-2 are maintained at 50 °C to facilitate effective dissociation of the hydrates. In the first stage, CO₂ captured and decomposed in D-1 is preserved for potential use in other processes. In contrast, the CO₂ in the second stage (R-2), after meeting emission standards, is directly released into the atmosphere as tail gas.

4.1.2 The process of high-pressure hydrate formation and atmospheric dissociation (H-A)

Figure 6 illustrates a process flow designed to maintain pressurized dissociation. In that configuration, each hydrate formation reactor and decomposition reactor require a dedicated pump to manage pressure levels. However, in this section, an atmospheric dissociation strategy is adopted. The resulting process flow is depicted in Fig. 7.

Compared to pressurized dissociation, the major advantage of operating under atmospheric conditions is that the decomposition reactors require less thermal input. Nonetheless, a considerable pressure difference arises between the atmospheric decomposition reactor and the subsequent, higher-pressure formation reactor. Consequently, pumps (P-2 and P-3) must overcome a large pressure gap, leading to increased energy consumption. Because a high-pressure formation pathway is still employed, the left-hand segment of Fig. 7 is essentially the same as in Fig. 6.

Overall, this approach demonstrates the trade-offs between reduced thermal requirements in the decomposition stage and the higher electrical power demand for pumping. Such design considerations guide the selection of optimal operating pressures in hydrate-based processes, balancing both energy consumption and equipment complexity.

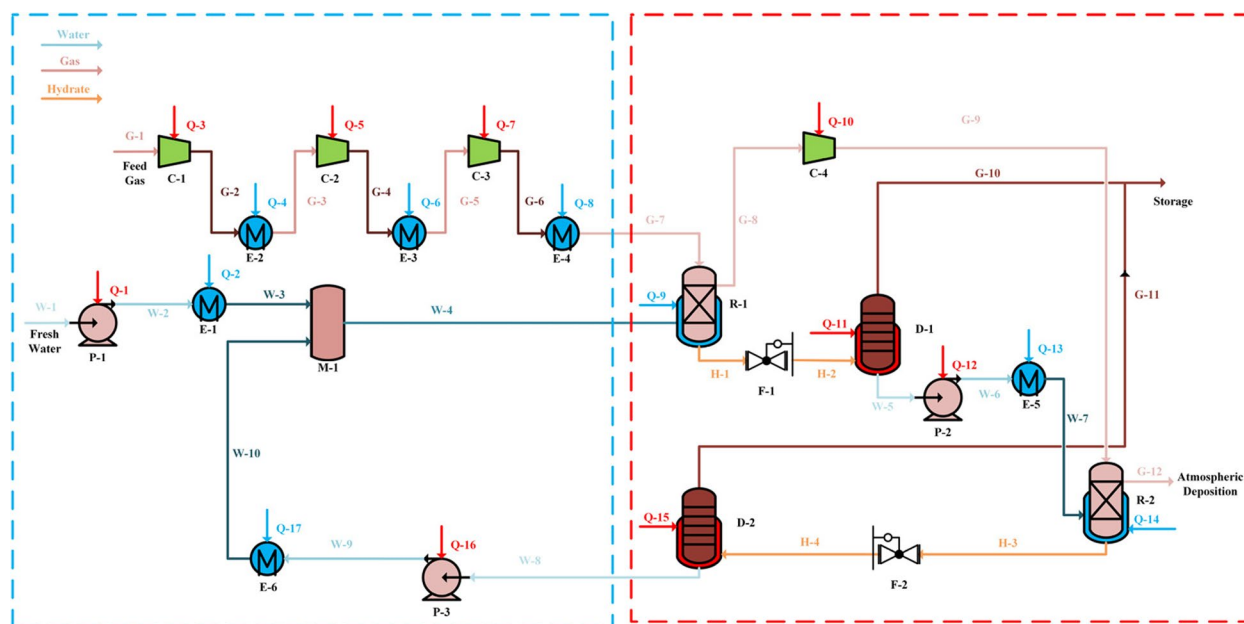


Fig. 7 The process flow diagram of high-pressure hydrate formation and atmospheric dissociation

4.1.3 The process of low-pressure hydrate formation and pressured dissociation (L-P)

According to the experimental results presented in Fig. 4, operating under a relatively low formation pressure (1.5 MPa) requires a three-stage hydrate formation and dissociation to achieve an equivalent level of CO₂ removal. Building on these findings, a corresponding process flow diagram has been established, as illustrated in Fig. 8.

In this process, because the hydrate is formed at 1.5 MPa—a lower pressure compared to the high-pressure scheme shown in Fig. 6—the feed gas (G-1) requires only one compression step (C-1) and a subsequent cooling step (E-2) to reach the operating conditions. The key difference lies in the right-hand portion of Fig. 8, where three consecutive hydrate formation and decomposition stages are employed: namely, (R-1, D-1), (R-2, D-2), and (R-3, D-3). This multi-stage arrangement is designed to achieve comparable decarbonization performance at lower hydrate formation pressures.

4.1.4 The process of low-pressure hydrate formation and atmospheric dissociation (L-A)

Similar to the H-A process, the low-pressure configuration also incorporates atmospheric-pressure dissociation in the hydrate dissociation reactor, as illustrated in Fig. 9.

Compared with Fig. 8, the key change is that the pumps originally placed between the hydrate formation reactor and the dissociation reactor at each stage are replaced with pressure-reducing valves (F-1, F-2, and F-3). Meanwhile, pumps P-2, P-3, and P-4 are employed to raise the

pressure of water—after it has been depressurized for dissociation—back to the reaction pressure of 1.5 MPa.

4.2 4E analysis of hydrate-based CO₂ capture process

Although these processes (H-P, H-A, L-P, and L-A) differ in their operating pressures and dissociation conditions, they all aim to achieve effective CO₂ capture via hydrate formation and decomposition. To identify the most suitable approach for industrial application, a systematic assessment of energy consumption, exergy efficiency, environmental impact, and economic feasibility (the so-called 4E analysis) is required. The following section (Sect. 4.2) presents the 4E analysis for each process, offering insights into their respective advantages, limitations, and overall viability.

4.2.1 Energy analysis

Building upon the detailed process descriptions in Sect. 4.1, it is crucial to evaluate the energy performance of each hydrate-based CO₂ capture process to determine their feasibility and efficiency. The energy analysis compares the total energy consumption of the four distinct processes: High-Pressure Formation with Pressurized Dissociation (H-P), High-Pressure Formation with Atmospheric Dissociation (H-A), Low-Pressure Formation with Pressurized Dissociation (L-P), and Low-Pressure Formation with Atmospheric Dissociation (L-A).

Figure 10 compares the total energy consumption of the four investigated processes—H-P, H-A, L-P, and L-A—and provides a breakdown by major equipment (heater, cooler, compressor/pump, formation reactor,

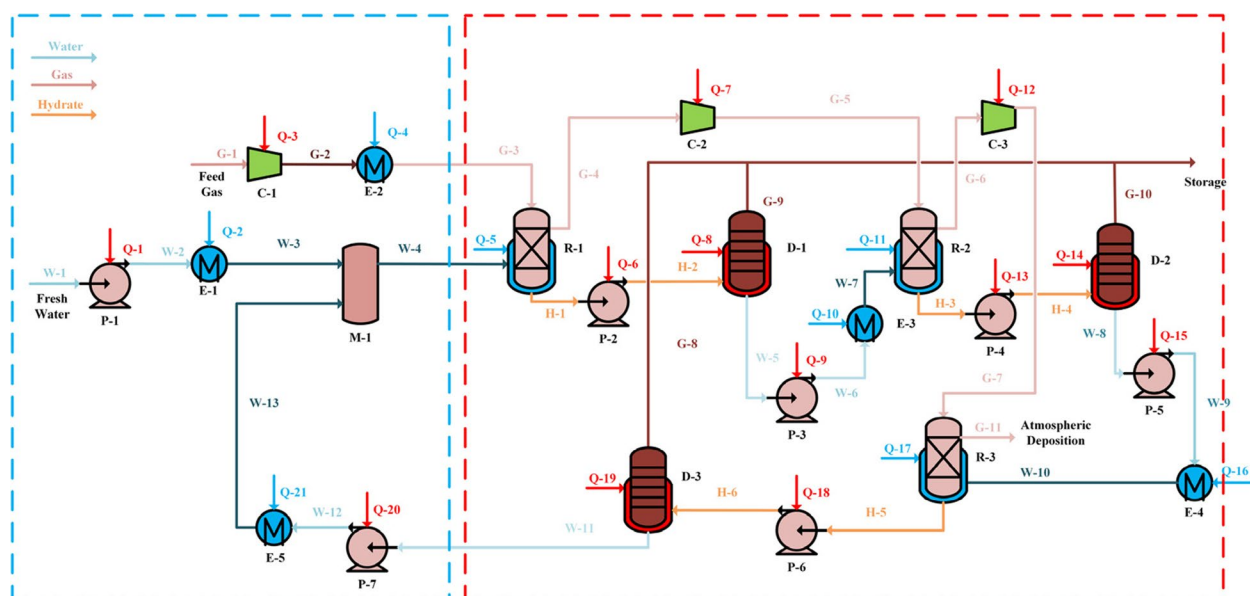


Fig. 8 The process flow diagram of low-pressure hydrate formation and pressured dissociation

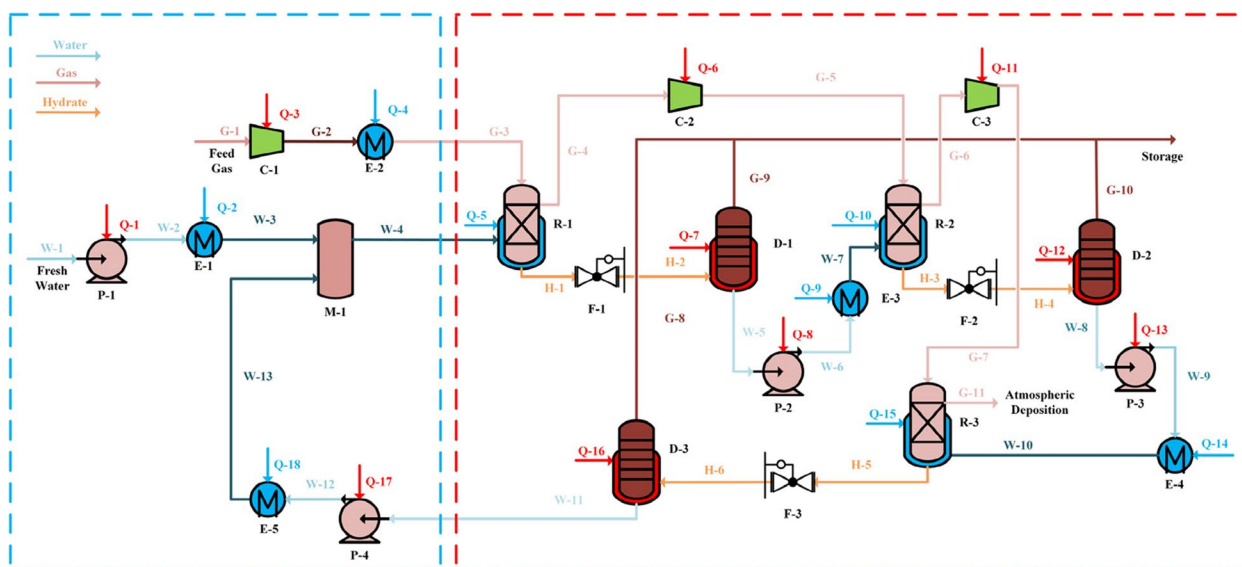


Fig. 9 The process flow diagram of low-pressure hydrate formation and atmospheric dissociation

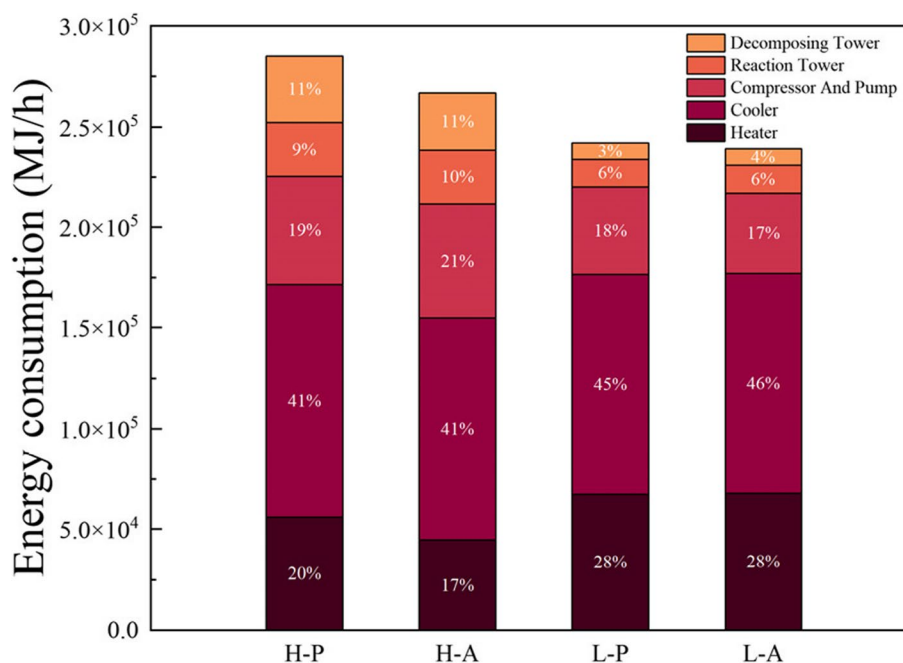


Fig. 10 Energy consumption of hydrate-based CO₂ capture processes (H-P, H-A, L-P and L-A)

and dissociation reactor). From left to right, the corresponding total energy demands are 285,122 MJ/h (H-P), 269,710 MJ/h (H-A), 246,587 MJ/h (L-P), and 240,076 MJ/h (L-A). The H-P process exhibits the highest total energy usage at approximately 285,123 MJ/h, primarily due to the substantial energy required for maintaining high pressures during both hydrate formation and dissociation. Transitioning to the H-A process

results in a notable reduction in energy consumption to 269,710 MJ/h, reflecting a decrease of about 5.4%. This improvement is mainly attributed to lower heating demands associated with atmospheric dissociation, despite a slight increase in energy required for pumping to manage the pressure differential.

Further energy savings are achieved with the L-P process, which operates at a reduced formation pressure

of 1.5 MPa, lowering the total energy consumption to 246,587 MJ/h. This represents a significant reduction of approximately 13% compared to the H-P process. The decrease is largely due to the reduced energy needed for gas compression at lower pressures, although some energy is still consumed in maintaining pressurized dissociation.

The most energy-efficient configuration is the L-A process, which combines low-pressure hydrate formation with atmospheric dissociation, resulting in the lowest total energy consumption of 240,077 MJ/h. This marks a 16.6% reduction from the H-P process. The L-A process benefits from both minimized compression energy and reduced thermal input for dissociation, despite a slight increase in cooling requirements due to the multi-stage formation process.

4.2.2 Exergy analysis

Following the comprehensive energy assessment in Sect. 4.2.1, it is imperative to delve into the exergy performance of each hydrate-based CO₂ capture process to gain deeper insights into their thermodynamic efficiencies and identify areas of potential improvement. Exergy analysis in Fig. 11 provides a more nuanced evaluation by accounting for the quality of energy and the irreversibility within each process, thereby complementing the energy consumption metrics previously discussed.

Figure 11 presents the total exergy efficiencies of the four processes: H-P, H-A, L-P, and L-A. The exergy efficiency trends align closely with the energy consumption analysis, where the H-P process exhibits the lowest

exergy efficiency of 0.607 due to the substantial irreversibility associated with maintaining high pressures during both hydrate formation and dissociation. Conversely, the L-A process demonstrates the highest exergy efficiency of 0.725, benefiting from reduced compression and thermal requirements.

Examining the exergy flow diagrams for the H-P and L-P processes, illustrated in Figs. 12 and 13 respectively, reveals the primary sources of exergy destruction. In the H-P process, significant exergy losses occur in the compression stages and heat exchangers, where high-pressure operations and temperature gradients introduce considerable irreversibility. The dissociation reactors, operating under pressurized conditions, further contribute to exergy destruction through the required thermal inputs. In contrast, the L-P process, operating at a lower formation pressure of 1.5 MPa, shows reduced exergy losses in compression. However, some exergy destruction still arises from the pressurized dissociation stage, albeit to a lesser extent than in H-P.

Comparative analysis between H-P and H-A processes, as depicted in SI Fig. 2, highlights the impact of shifting from pressurized to atmospheric dissociation. The H-A process exhibits improved exergy efficiency relative to H-P, primarily due to lower thermal inputs required for atmospheric dissociation. This transition mitigates some of the exergy losses associated with high-temperature operations. Nevertheless, the increased energy demand for pumping to overcome the pressure differential introduces new sources of exergy destruction, albeit these are offset by the reductions in thermal inefficiencies.

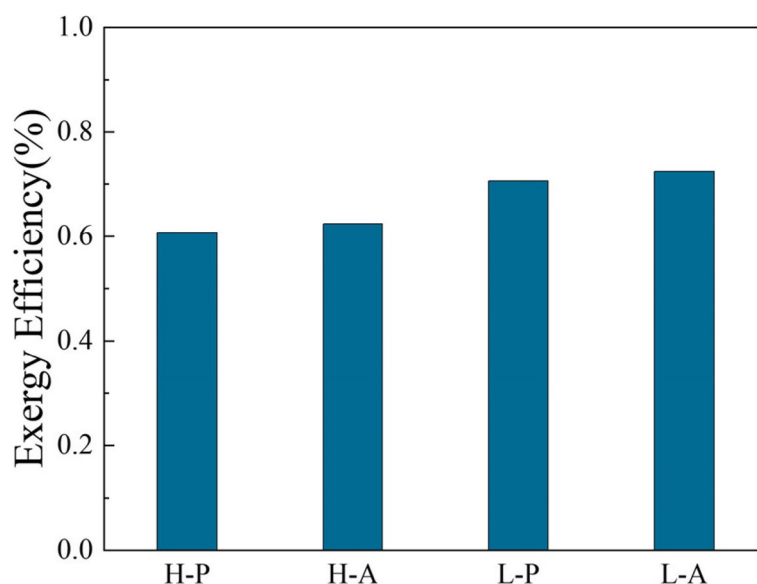


Fig. 11 Exergy efficiency of hydrate-based CO₂ capture processes (H-P, H-A, L-P and L-A)

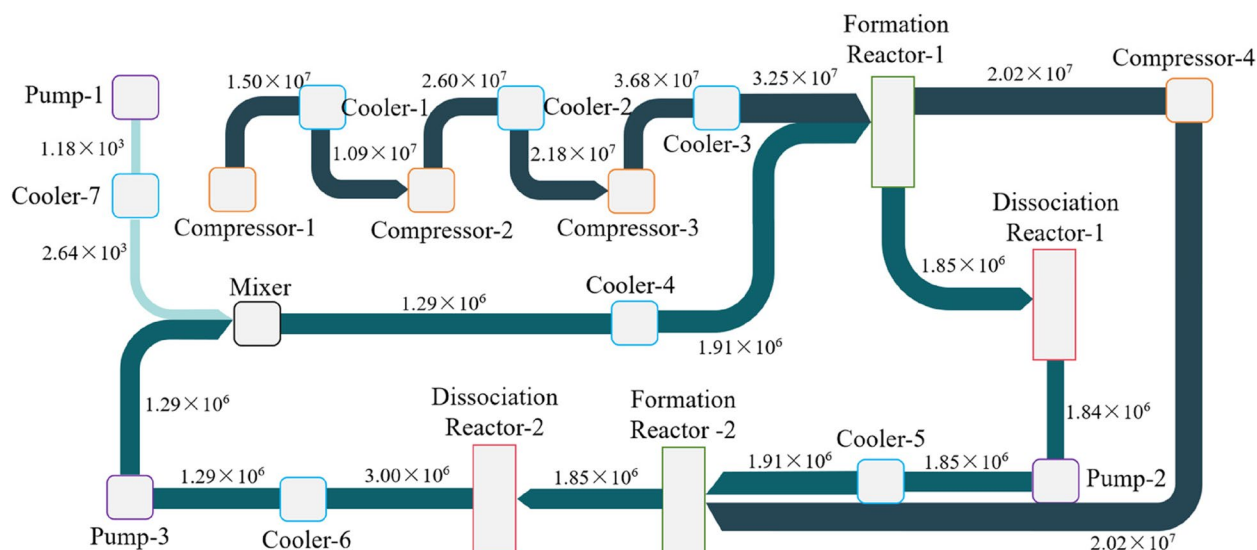


Fig. 12 Exergy flow diagram of the H-P hydrate-based CO₂ capture process

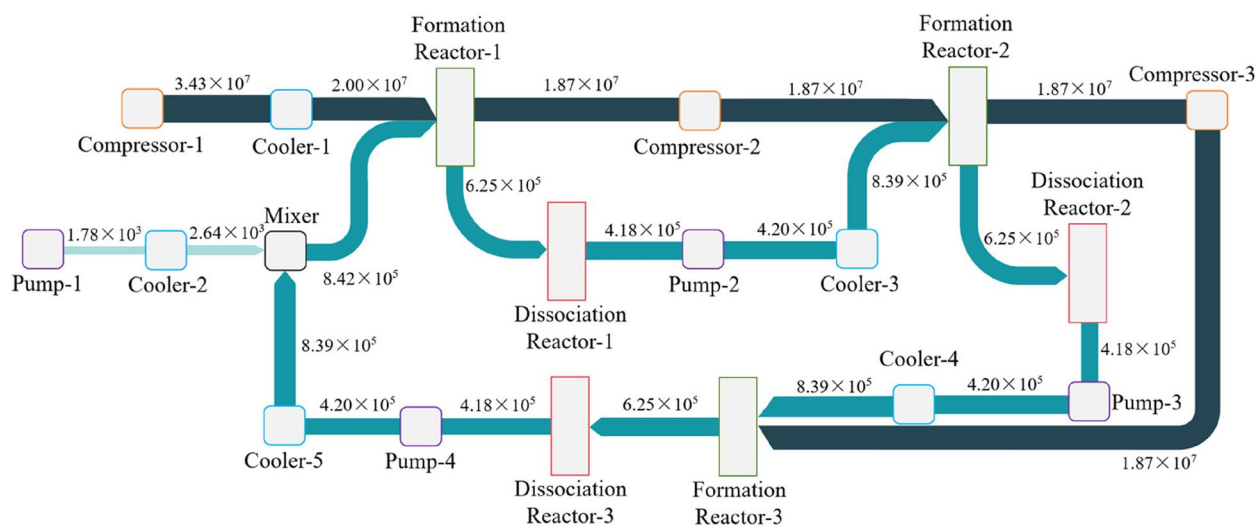


Fig. 13 Exergy flow diagram of the L-P hydrate-based CO₂ capture process

Similarly, the comparison between L-P and L-A processes in SI Fig. 3 underscores the benefits of integrating low-pressure formation with atmospheric dissociation. The L-A process not only minimizes exergy losses in compression by operating at reduced formation pressures but also capitalizes on the lower thermal requirements of atmospheric dissociation. This dual optimization leads to a cumulative enhancement in exergy efficiency, positioning the L-A process as the most thermodynamically favorable configuration among the evaluated options.

Overall, the exergy analysis corroborates the energy consumption findings, reinforcing the superiority of the

L-A process in terms of thermodynamic efficiency. By lowering the formation pressure and adopting atmospheric dissociation, the L-A configuration effectively reduces both compression and thermal exergy losses, thereby enhancing the overall exergy efficiency. This makes the L-A process not only the most energy-efficient but also the most exergy-efficient strategy for hydrate-based CO₂ capture.

4.2.3 Economic analysis

Building upon the energy and exergy assessments in the preceding sections, it is crucial to evaluate the

economic viability of each hydrate-based CO₂ capture process to determine their practical applicability and cost-effectiveness. The economic analysis considers both the total equipment costs and the operational costs per cubic meter of raw gas processed for the four distinct processes.

The total equipment costs in Fig. 14 reveal a significant disparity between high-pressure and low-pressure processes. The H-P and H-A configurations incur substantially higher equipment investments, with costs of 12.78585 million USD and 13.254 million USD, respectively. This elevated expense is primarily attributable to the infrastructure required to sustain high-pressure operations, including specialized compressors and reactors. In contrast, the L-P and L-A processes exhibit markedly lower equipment costs, amounting to 6.54341 million USD and 6.80332 million USD, respectively. The reduced financial burden of low-pressure systems stems from simpler and less costly equipment, as well as decreased requirements for high-strength materials.

When examining the operational costs in Fig. 15, specifically the cost per cubic meter of raw gas processed, a similar trend emerges. The H-P and H-A processes incur higher costs of 0.0251 USD/m³ and 0.0249 USD/m³, respectively, reflecting the increased energy consumption and maintenance associated with high-pressure operations. Conversely, the L-P and L-A configurations demonstrate more economical processing costs of 0.0206 USD/m³ and 0.0198 USD/m³, respectively. These lower operational expenses are a direct consequence of reduced energy demands, as highlighted in the energy analysis,

and the streamlined process design inherent to low-pressure systems.

Overall, the economic analysis underscores the financial advantages of low-pressure hydrate formation processes, both in terms of initial capital investment and ongoing operational costs. The L-A process emerges as the most cost-effective option, offering the lowest equipment and processing costs among the evaluated configurations. This economic superiority, combined with the favorable energy and exergy performance previously discussed, positions the L-A process as a highly attractive solution for industrial-scale CO₂ capture applications.

4.2.4 Environmental analysis

Building upon the energy, exergy, and economic evaluations presented in the preceding sections, it is imperative to assess the environmental implications of each hydrate-based CO₂ capture process. This analysis in Fig. 16 focuses on the indirect CO₂ emissions associated with the removal of each kilogram of CO₂, providing a critical measure of the environmental sustainability of the processes. Specifically, the ratio of indirect CO₂ emissions to the amount of CO₂ removed serves as an indicator: values exceeding one signify that the process generates more CO₂ indirectly than it captures, rendering it environmentally detrimental or neutral, whereas values below one indicates a net environmental benefit.

The H-P and H-A processes exhibit indirect CO₂ emission ratios greater than one, highlighting their environmental challenges. These elevated ratios are primarily attributable to the substantial energy and material requirements

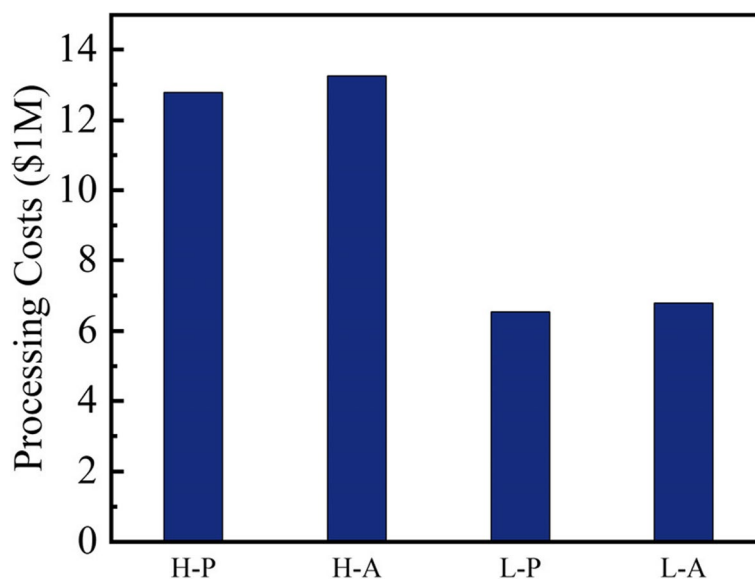


Fig. 14 The total equipment costs of hydrate-based CO₂ capture processes (H-P, H-A, L-P and L-A)

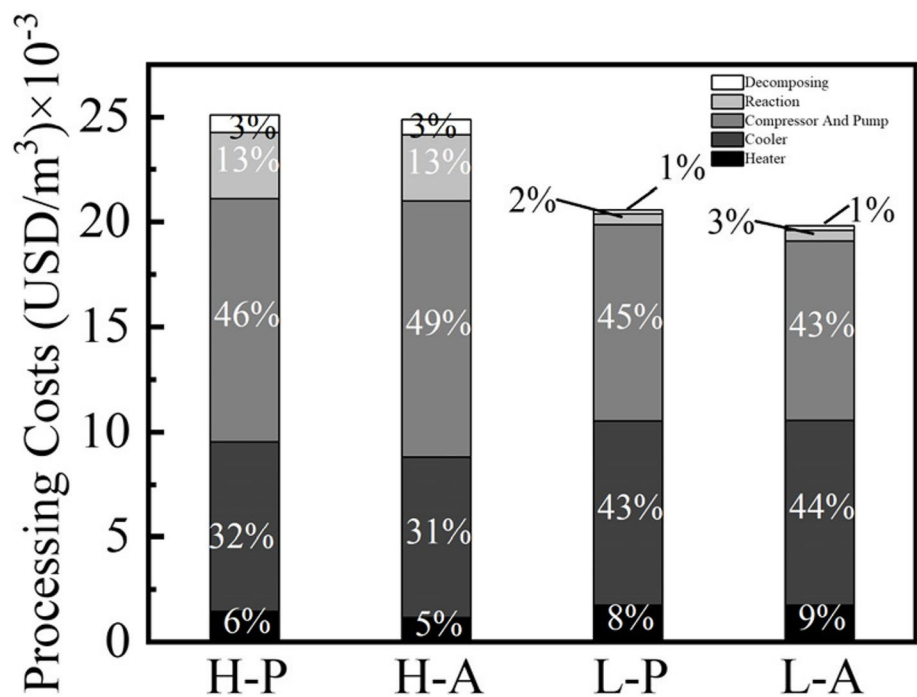


Fig. 15 The operational costs of hydrate-based CO₂ capture processes (H-P, H-A, L-P and L-A)

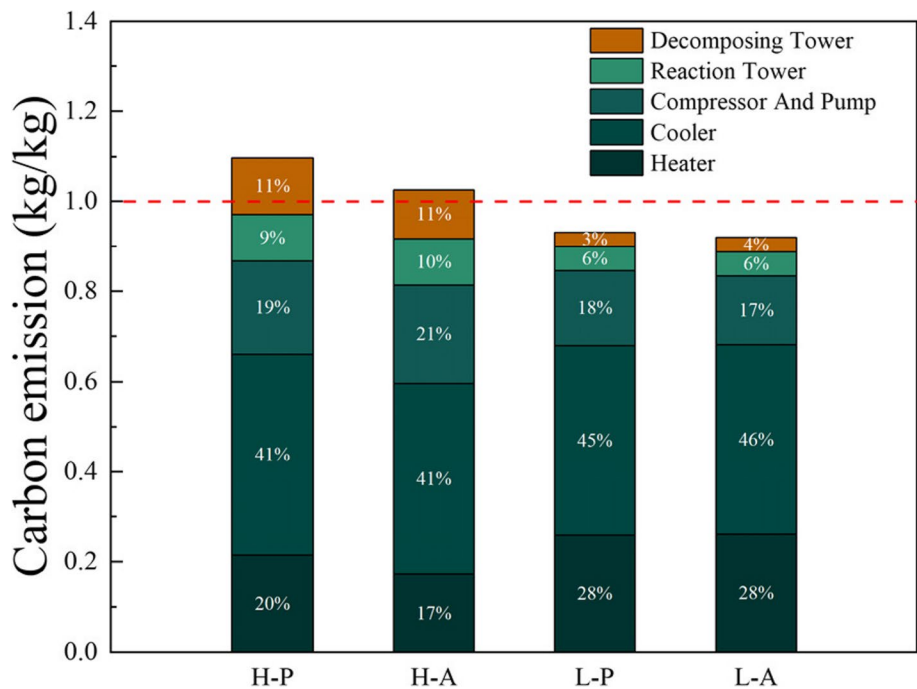


Fig. 16 The carbon emission of hydrate-based CO₂ capture processes (H-P, H-A, L-P and L-A)

inherent in maintaining high-pressure conditions and the associated infrastructure demands. Specifically, the H-P process, with its dual reliance on high-pressure formation

and pressurized dissociation, results in the highest indirect emissions, reflecting the intensive energy consumption and the consequent CO₂ emissions from energy generation.

In contrast, the L-P and L-A processes demonstrate more favorable environmental profiles, with indirect CO₂ emission ratios below one. This improvement is largely due to the reduced formation pressures, which lower the energy demands for gas compression and thermal management. The L-P process benefits from operating at a significantly lower pressure of 1.5 MPa, which diminishes the indirect CO₂ emissions compared to its high-pressure counterparts. Moreover, the transition to atmospheric dissociation in the L-A process further enhances its environmental performance by eliminating the need for energy-intensive pressurization during the decomposition stage. Consequently, the L-A process not only minimizes indirect CO₂ emissions but also leverages the efficiencies gained from lower operational pressures, resulting in a net positive environmental impact.

In summary, the environmental analysis highlights a clear preference for low-pressure hydrate formation combined with atmospheric dissociation.

5 Conclusion

This study systematically evaluated hydrate-based CO₂ capture technologies for flue gas decarbonization through experiments and comprehensive process simulations. Based on experimental results, a two-stage high-pressure formation process with sodium p-styrenesulfonate as the kinetic promoter achieved a CO₂ removal rate of 53.65%, reducing the CO₂ concentration from 20 mol% to 9.27 mol%. In comparison, a three-stage low-pressure formation process using both kinetic and thermodynamic promoters resulted in a higher CO₂ removal rate of 64.66%, lowering the CO₂ concentration to 9.11 mol%.

Complementary process simulations and a detailed 4E (Energy, Exergy, Economy, and Environment) analysis identified the Low-Pressure Formation with Atmospheric Dissociation (L-A) configuration as the most effective approach. The L-A process achieved the lowest total energy consumption of 240,077 MJ/h and the highest exergy efficiency of 0.725. Economically, it presented significantly lower equipment and operational costs compared to high-pressure alternatives. Environmentally, the L-A configuration maintained indirect CO₂ emission ratios below one, indicating a net positive impact. These findings collectively suggest that the L-A process offers a balanced and efficient solution for industrial-scale CO₂ capture, combining technical feasibility, cost-effectiveness, and environmental sustainability. Consequently, the L-A configuration emerges as a promising candidate for advancing hydrate-based gas separation technologies in mitigating industrial carbon emissions sustainably.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s43979-025-00125-y>.

Supplementary Material 1. SI Table 1 is the key parameters for the hypothetical CO₂ hydrate component. SI Fig. 1 is the CO₂ hydrate formation and dissociation reactor model. SI Fig. 2 is the comparative exergy analysis between H-P and H-A processes. SI Fig. 3 is the comparative exergy analysis between L-P and L-A processes.

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Authors' contributions

Xiaoming Wang: Conceptualization, Writing – original draft, Investigation; Shangjie Shao: Methodology, Formal analysis; Guangyong Yang: Visualization; Qixian Yan: Writing – original draft; Haoyu Yuan: Methodology, Chen Chen: Resources, Funding acquisition, Writing – Supervision; Fei Wang: Project administration, Funding acquisition.

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Data availability

Data and materials will be made available on reasonable request.

Declarations

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Adam L, Ballard (2002): A Non-Ideal Hydrate Solid Solution Model for a Multi-Phase Equilibria Program. Colorado School of Mines.
- Adeyemo A, Kumar R, Linga P, Ripmeester J, Englezos P (2010) Capture of carbon dioxide from flue or fuel gas mixtures by clathrate crystallization in a silica gel column. In International Journal of Greenhouse Gas Control 4(3):478–485. <https://doi.org/10.1016/j.jggc.2009.11.011>
- Aljehani A, Razack SA, K., Nitsche, Ludwig; Al-Hallaj, Said, (2018) Design and optimization of a hybrid air conditioning system with thermal energy storage using phase change composite. In Energy Conversion and Management 169:404–418. <https://doi.org/10.1016/j.enconman.2018.05.040>
- Al-Zareer M, Dincer I, Rosen MA (2017) Performance analysis of a supercritical water-cooled nuclear reactor integrated with a combined cycle, a Cu-Cl thermochemical cycle and a hydrogen compression system. In Applied Energy 195:646–658. <https://doi.org/10.1016/j.apenergy.2017.03.046>
- Aminnaji M, Qureshi M, Fahed; Dashti, Hossein; Hase, Alfred; Mosalanejad, Abdolali; Jahanbakhsh, Amir, et al (2024) CO₂ Gas hydrate for carbon capture and storage applications – Part 1. In Energy 300:131579. <https://doi.org/10.1016/j.energy.2024.131579>
- Aminnaji M, Qureshi M, Fahed; Dashti, Hossein; Hase, Alfred; Mosalanejad, Abdolali; Jahanbakhsh, Amir, et al (2024) CO₂ gas hydrate for carbon capture and storage applications – Part 2. In Energy 300:131580. <https://doi.org/10.1016/j.energy.2024.131580>

7. Bai D, Zhang D, Zhang X, Chen G (2015) Origin of Self-preservation Effect for Hydrate Decomposition: Coupling of Mass and Heat Transfer Resistances. In *Scientific reports* 5:14599. <https://doi.org/10.1038/srep14599>
8. Bai J, Cheng C, Wei Y, Yan K, Li P, Fang S, Chang C (2021) Thermodynamic investigation of hydrate-based CO₂ capture from simulated flue gas with new mixed promoters. In *International Journal of Chemical Reactor Engineering* 19(1):75–85. <https://doi.org/10.1515/ijcre-2020-0147>
9. Bavoh CB, Lal B, Osei H, Sabli KM, Mukhtar H (2019) A review on the role of amino acids in gas hydrate inhibition, CO₂ capture and sequestration, and natural gas storage. In *Journal of Natural Gas Science and Engineering* 64:52–71. <https://doi.org/10.1016/j.jngse.2019.01.020>
10. Berstad D, Anantharaman R, Nekså P (2013) Low-temperature CO₂ capture technologies – Applications and potential. In *International Journal of Refrigeration* 36(5):1403–1416. <https://doi.org/10.1016/j.jirefrig.2013.03.017>
11. Chen C, Yuan H, Bi R, Wang Na, Li Y, He Y, Wang F (2022) A novel conceptual design of LNG-sourced natural gas peak-shaving with gas hydrates as the medium. In *Energy* 253:124169. <https://doi.org/10.1016/j.energy.2022.124169>
12. Dhamu V, Mengqi X, Qureshi M, Fahed; Yin, Zhenyuan; Jana, Amiya K., Linga, Praveen, (2024) Evaluating CO₂ hydrate kinetics in multi-layered sediments using experimental and machine learning approach: Applicable to CO₂ sequestration. In *Energy* 290:129947. <https://doi.org/10.1016/j.energy.2023.129947>
13. Dhamu V, Qureshi M, Fahed; Abubakar, Saifudin; Usadi, Adam; Barckholtz, Timothy A., Mhadeshwar, Ashish B., Linga, Praveen, (2023) Investigating High-Pressure Liquid CO₂ Hydrate Formation, Dissociation Kinetics, and Morphology in Brine and Freshwater Static Systems. In *Energy Fuels* 37(12):8406–8420. <https://doi.org/10.1021/acs.energyfuels.3c01089>
14. Dhamu V, Qureshi M, Fahed; Barckholtz, Timothy A., Mhadeshwar, Ashish B., Linga, Praveen, (2023) Evaluating liquid CO₂ hydrate formation kinetics, morphology, and stability in oceanic sediments on a lab scale using top injection. In *Chemical Engineering Journal* 478:147200. <https://doi.org/10.1016/j.cej.2023.147200>
15. Dhamu V, Xiao M, Qureshi M, Fahed; Linga, Praveen, (2024) Deciphering the CO₂ hydrates formation dynamics in brine-saturated oceanic sediments using experimental and machine learning modelling approach. In *Energy* 313:133802. <https://doi.org/10.1016/j.energy.2024.133802>
16. E. Dendy Sloan Jr.; Carolyn A. Koh (2007): *Clathrate Hydrates of Natural Gases*: CRC Press. Available online at <https://www.taylorfrancis.com/books/mono/10.1201/9781420008494/clathrate-hydrates-natural-gases-dendy-sloan-jr-carolyn-koh-carolyn-koh>.
17. Fahed Qureshi M, Zheng J, Khandelwal H, Venkataraman P, Usadi A, Barckholtz TA et al (2022) Laboratory demonstration of the stability of CO₂ hydrates in deep-oceanic sediments. In *Chemical Engineering Journal* 432:134290. <https://doi.org/10.1016/j.cej.2021.134290>
18. Fan R, Xi H (2022) Energy, exergy, economic (3E) analysis, optimization and comparison of different Carnot battery systems for energy storage. In *Energy Conversion and Management* 252:115037. <https://doi.org/10.1016/j.enconman.2021.115037>
19. Fang M, Yi N, Di W, Wang T, Wang Q (2020) Emission and control of flue gas pollutants in CO₂ chemical absorption system – A review. In *International Journal of Greenhouse Gas Control* 93:102904. <https://doi.org/10.1016/j.ijggc.2019.102904>
20. Gambelli A, Maria; Presciutti, Andrea; Rossi, Federico, (2021) Review on the characteristics and advantages related to the use of flue-gas as CO₂/N₂ mixture for gas hydrate production. In *Fluid Phase Equilibria* 541:113077. <https://doi.org/10.1016/j.fluid.2021.113077>
21. Hashimoto H, Yamaguchi T, Kinoshita T, Muromachi S (2017) Gas separation of flue gas by tetra- n -butylammonium bromide hydrates under moderate pressure conditions. In *Energy* 129:292–298. <https://doi.org/10.1016/j.energy.2017.04.074>
22. He Y, Wang F (2018) Hydrate-based CO₂ capture: kinetic improvement via graphene-carried –SO₃– and Ag nanoparticles. In *J Mater Chem A* 6(45):22619–22625. <https://doi.org/10.1039/c8ta08785g>
23. Hou R, Fong C, Freeman BD, Hill MR, Xie Z (2022) Current status and advances in membrane technology for carbon capture. In *Separation and Purification Technology* 300:121863. <https://doi.org/10.1016/j.seppur.2022.121863>
24. Jeenuang K, Pornaroontham P, Fahed Qureshi M, Linga P, Rangsunvigit P (2024) Micro kinetic analysis of the CO₂ hydrate formation and dissociation with L-tryptophan in brine via high pressure in situ Raman spectroscopy for CO₂ sequestration. In *Chemical Engineering Journal* 479:147691. <https://doi.org/10.1016/j.cej.2023.147691>
25. Karimi, Mohsen; Shirzad, Mohammad; Silva, José A. C.; Rodrigues, Alirio E. (2023): Carbon dioxide separation and capture by adsorption: a review. In *Environmental chemistry letters*, pp. 1–44. <https://doi.org/10.1007/s10311-023-01589-z>.
26. Khaleel OJ, Ismail B, Firas; Khalil Ibrahim, Tamir; bin Abu Hassan, Saiful Hasmady, (2022) Energy and exergy analysis of the steam power plants: A comprehensive review on the Classification, Development, Improvements, and configurations. In *Ain Shams Engineering Journal* 13(3):101640. <https://doi.org/10.1016/j.jasej.2021.11.009>
27. Khalilpour R, Mumford K, Zhai H, Abbas A, Stevens G, Rubin ES (2015) Membrane-based carbon capture from flue gas: a review. In *Journal of Cleaner Production* 103:286–300. <https://doi.org/10.1016/j.jclepro.2014.10.050>
28. Khandelwal H, Qureshi M, Fahed; Zheng, Junjie; Venkataraman, Pradeep; Barckholtz, Timothy A., Mhadeshwar, Ashish B., Linga, Praveen, (2021) Effect of L-Tryptophan in Promoting the Kinetics of Carbon Dioxide Hydrate Formation. In *Energy Fuels* 35(1):649–658. <https://doi.org/10.1021/acs.energyfuels.0c03709>
29. Kim S, Lee S, Hyuk; Kang, Yong Tae, (2017) Characteristics of CO₂ hydrate formation/dissociation in H₂O + THF aqueous solution and estimation of CO₂ emission reduction by district cooling application. In *Energy* 120:362–373. <https://doi.org/10.1016/j.energy.2016.11.086>
30. Lee D, Lee Y, Lim J, Seo Y (2017) Guest enclathration and structural transition in CO₂+ N₂+ methylcyclopentane hydrates and their significance for CO₂ capture and sequestration. In *Chemical Engineering Journal* 320:43–49. <https://doi.org/10.1016/j.cej.2017.03.019>
31. Li G, Liu D, Xie Y, Xiao Y (2010) Study on Effect Factors for CO₂ Hydrate Rapid Formation in a Water-Spraying Apparatus. In *Energy Fuels* 24(8):4590–4597. <https://doi.org/10.1021/ef100417y>
32. Li, Kexin; Shi, Lingfeng; Zhang, Yonghao; Yao, Yu; Tian, Hua; Shu, Gequn (2024): Global selection appraisal study for heat pump system of electric vehicle based on energetic, economic, and environmental analysis. In *Carbon Neutrality* 3 (1). <https://doi.org/10.1007/s43979-024-00091-x>.
33. Li X, Zhang Y, Fang L, Jin Z, Zhang Y, Yu X et al (2019) Energy, exergy, economic, and environmental analysis of an integrated system of high-temperature heat pump and gas separation unit. In *Energy Conversion and Management* 198:111911. <https://doi.org/10.1016/j.enconman.2019.111911>
34. Liu X, Li Y, Chen G-J, Chen D-Y, Sun Bo, Yin Z (2023) Coupling Amino Acid with THF for the Synergistic Promotion of CO₂ Hydrate Micro Kinetics: Implication for Hydrate-Based CO₂ Sequestration. In *ACS Sustainable Chem Eng* 11(15):6057–6069. <https://doi.org/10.1021/acssuschemeng.3c00593>
35. Liu X, Ren J, Chen D, Yin Z (2022) Comparison of SDS and L-Methionine in promoting CO₂ hydrate kinetics: Implication for hydrate-based CO₂ storage. In *Chemical Engineering Journal* 438:135504. <https://doi.org/10.1016/j.cej.2022.135504>
36. Liu Y, Bisson TM, Yang H, Xu Z (2010) Recent developments in novel sorbents for flue gas clean up. In *Fuel Processing Technology* 91(10):1175–1197. <https://doi.org/10.1016/j.fuproc.2010.04.015>
37. Ma S, Tian X, Liu Z, Wu Z, Li G, Guan X et al (2024) Formation and decomposition characteristics of CO₂+TBAB hydrate for a safer CO₂ storage. In *Energy* 307:132801. <https://doi.org/10.1016/j.energy.2024.132801>
38. Ma, Xudong; Wu, Yuting; Du, Yanjun; Zhang, Cancan; Lei, Biao; Lu, Yuanwei (2024b): Comprehensive performance evaluation of steam generating heat pumps for industrial waste heat recovery. In *Carbon Neutrality* 3 (1). <https://doi.org/10.1007/s43979-024-00111-w>.
39. Ma ZW, Zhang P, Bao HS, Deng S (2016) Review of fundamental properties of CO₂ hydrates and CO₂ capture and separation using hydration method. In *Renewable and Sustainable Energy Reviews* 53:1273–1302. <https://doi.org/10.1016/j.rser.2015.09.076>
40. Mei B, Barnoon P, Toghrade D, Su C-H, Nguyen H, Chinh; Khan, Afrasyab, (2022) Energy, exergy, environmental and economic analyzes (4E) and multi-objective optimization of a PEM fuel cell equipped with coolant channels. In *Renewable and Sustainable Energy Reviews* 157:112021. <https://doi.org/10.1016/j.rser.2021.112021>
41. Mu L, Wang S, Lu J, Li C, Lan Y, Liu G, Zhang T (2024) Effect of hydrogen-enriched natural gas on flue gas waste heat recovery potential and

- condensing heat exchanger performance. In *Energy* 286:129591. <https://doi.org/10.1016/j.energy.2023.129591>
42. Mu L, Wang S, Lu J, Liu G, Zhao L, Lan Y (2023) Effect of flue gas condensing waste heat recovery and its pressure drop on energy saving and carbon reduction for refinery heating furnace. In *Energy* 279:128081. <https://doi.org/10.1016/j.energy.2023.128081>
 43. Pasichnyk M, Stanovsky P, Polezhaev P, Zach B, Šyc M, Bobák M et al (2023) Membrane technology for challenging separations: Removal of CO₂, SO₂ and NO_x from flue and waste gases. In *Separation and Purification Technology* 323:124436. <https://doi.org/10.1016/j.seppur.2023.124436>
 44. Phan A, Schlösser H, Striolo A (2021) Molecular mechanisms by which tetrahydrofuran affects CO₂ hydrate Growth: Implications for carbon storage. In *Chemical Engineering Journal* 418:129423. <https://doi.org/10.1016/j.cej.2021.129423>
 45. Qureshi M, Fahed, Mert; Altamash, Tausif; Aparicio, Santiago; Aminnaji, Morteza; Tohidi, Bahman, (2017) High-pressure gas hydrate autoclave hydraulic experiments and scale-up modeling on the effect of stirring RPM effect. In *Journal of Natural Gas Science and Engineering* 38:50–58. <https://doi.org/10.1016/j.jngse.2016.12.023>
 46. Qureshi M, Fahed; Dhamu, Vikas; Usadi, Adam; Barckholtz, Timothy A., Mhadeshwar, Ashish B., Linga, Praveen, (2022) CO₂ Hydrate Formation Kinetics and Morphology Observations Using High-Pressure Liquid CO₂ Applicable to Sequestration. In *Energy Fuels* 36(18):10627–10641. <https://doi.org/10.1021/acs.energyfuels.1c03840>
 47. Qureshi M, Fahed; Khandelwal, Himanshu; Usadi, Adam; Barckholtz, Timothy A., Mhadeshwar, Ashish B., Linga, Praveen, (2022) CO₂ hydrate stability in oceanic sediments under brine conditions. In *Energy* 256:124625. <https://doi.org/10.1016/j.energy.2022.124625>
 48. Shoaie M, Hajinezhad A, Moosavian SF (2023) Design, energy, exergy, economy, and environment (4E) analysis, and multi-objective optimization of a novel integrated energy system based on solar and geothermal resources. In *Energy* 280:128162. <https://doi.org/10.1016/j.energy.2023.128162>
 49. Sloan ED (2003) Fundamental principles and applications of natural gas hydrates. In *Nature* 426(6964):353–363. <https://doi.org/10.1038/nature02135>
 50. Solomon, Susan (2007): Climate change 2007. The physical science basis contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge, New York: Cambridge University Press.
 51. Sum AK, Koh CA, Sloan ED (2009) Clathrate Hydrates: From Laboratory Science to Engineering Practice. In *Ind Eng Chem Res* 48(16):7457–7465. <https://doi.org/10.1021/ie900679m>
 52. Sun Q, Kang YT (2015) Experimental correlation for the formation rate of CO₂ hydrate with THF (tetrahydrofuran) for cooling application. In *Energy* 91:712–719. <https://doi.org/10.1016/j.energy.2015.08.089>
 53. Sun Q, Kang YT (2016) Review on CO₂ hydrate formation/dissociation and its cold energy application. In *Renewable and Sustainable Energy Reviews* 62:478–494. <https://doi.org/10.1016/j.rser.2016.04.062>
 54. Takeya S, Muromachi S, Yamamoto Y, Umeda H, Matsuo S (2016) Preservation of CO₂ hydrate under different atmospheric conditions. In *Fluid Phase Equilibria* 413:137–141. <https://doi.org/10.1016/j.fluid.2015.10.036>
 55. Tegze G, Gránásky L, Kvamme B (2007) Phase field modeling of CH₄ hydrate conversion into CO₂ hydrate in the presence of liquid CO₂. In *Physical chemistry chemical physics : PCCP* 9(24):3104–3111. <https://doi.org/10.1039/B700423K>
 56. Towler, Gavin; Sinnott, Ray: Design of Pressure Vessels, pp. 563–629. <https://doi.org/10.1016/B978-0-08-096659-5.00014-6>
 57. Vishwakarma G, Dhamu V, Qureshi M, Fahed; Bhattacharjee, Gaurav; Pradeep, Thalappil; Linga, Praveen, (2025) Understanding the Kinetics of CO₂ Hydrate Formation in Dry Water for Carbon Capture and Storage: X-ray Diffraction and In Situ Raman Studies. In *ACS applied materials & interfaces* 17(3):4865–4874. <https://doi.org/10.1021/acsami.4c17727>
 58. Walski TM (2012) Planning-Level Capital Cost Estimates for Pumping. In *J Water Resour Plann Manage* 138(3):307–310. [https://doi.org/10.1061/\(ASCE\)WR.1943-5452.0000167](https://doi.org/10.1061/(ASCE)WR.1943-5452.0000167)
 59. Wang F, Song Y-M, Liu G-Q, Guo G, Luo S-J, Guo R-B (2018) Rapid methane hydrate formation promoted by Ag&SDS-coated nanospheres for energy storage. In *Applied Energy* 213:227–234. <https://doi.org/10.1016/j.apene.2018.01.021>
 60. Wang X, Qiao L, Deng C, Chu Ge, Li X, Zhao Qi, Wang G (2021) Study on the characteristics of nitrogen dioxide adsorption and storage of coal residue in coal-fired power plants in goaf. In *Scientific reports* 11(1):8822. <https://doi.org/10.1038/s41598-021-87855-y>
 61. Wang Y, Lang X, Fan S (2013) Hydrate capture CO₂ from shifted synthesis gas, flue gas and sour natural gas or biogas. In *Journal of Energy Chemistry* 22(1):39–47. [https://doi.org/10.1016/S2095-4956\(13\)60004-2](https://doi.org/10.1016/S2095-4956(13)60004-2)
 62. Wiheeb AD, Helwani Z, Kim J, Othman MR (2016) Pressure Swing Adsorption Technologies for Carbon Dioxide Capture. In *Separation & Purification Reviews* 45(2):108–121. <https://doi.org/10.1080/15422119.2015.1047958>
 63. Yan He; Meng-Ting Sun; Chen Chen; Guo-Dong Zhang; Kun Chao; Yan Lin; Fei Wang, (2019) Surfactant-based promotion to gas hydrate formation for energy storage. In *J Mater Chem A* 7(38):21634–21661. <https://doi.org/10.1039/C9TA07071K>
 64. Yao Y, Yin Z, Niu M, Liu X, Zhang J, Chen D (2023) Evaluation of 1,3-dioxolane in promoting CO₂ hydrate kinetics and its significance in hydrate-based CO₂ sequestration. In *Chemical Engineering Journal* 451:138799. <https://doi.org/10.1016/j.cej.2022.138799>
 65. Yodpetch V, Inkong K, Veluswamy H, Prakash; Kulprathipanja, Santi; Rangsunvigit, Pramoch; Linga, Praveen, (2023) Investigation on the Amino Acid-Assisted CO₂ Hydrates: A Promising Step Toward Hydrate-based Decarbonization. In *ACS Sustainable Chem Eng* 11(7):2797–2809. <https://doi.org/10.1021/acssuschemeng.2c05967>
 66. Yu C-H, Huang C-H, Tan C-S (2012) A Review of CO₂ Capture by Absorption and Adsorption. In *Aerosol Air Qual Res* 12(5):745–769. <https://doi.org/10.4209/aaqr.2012.05.0132>
 67. Zhang G, Shen Y, Qi Y, Shan H, Li B, Xu L (2023) Selective Adsorption of CO₂ and N₂ during Hydrate Formation in Water-Rich Environments. In *Energy Fuels* 37(21):16648–16659. <https://doi.org/10.1021/acs.energyfuels.3c01708>
 68. Zhang G, Song P, Kong Y, Wang F (2023) The first trial of hydrate continuous production: Paving way on hydrate-based carbon capture and storage. In *Energy* 282:128917. <https://doi.org/10.1016/j.energy.2023.128917>
 69. Zhang H, Wang L, Lin X, Chen H (2023) Parametric optimisation and thermo-economic analysis of Joule-Brayton cycle-based pumped thermal electricity storage system under various charging–discharging periods. In *Energy* 263:125908. <https://doi.org/10.1016/j.energy.2022.125908>
 70. Zhang J, Li Y, Yin Z, Linga P, He T, Zheng XY (2023) Coupling amino acid L-Val with THF for superior hydrogen hydrate kinetics: Implication for hydrate-based hydrogen storage. In *Chemical Engineering Journal* 467:143459. <https://doi.org/10.1016/j.cej.2023.143459>
 71. Zhang Y, Chen F, He Y, Wang F (2022) Enhanced CO₂ hydrate formation via biopromoter coupled with initial stirring activation. In *Fuel* 330:125713. <https://doi.org/10.1016/j.fuel.2022.125713>
 72. Zheng J, Chong Z, Rong; Qureshi, M. Fahed; Linga, Praveen, (2020) Carbon Dioxide Sequestration via Gas Hydrates: A Potential Pathway toward Decarbonization. In *Energy Fuels* 34(9):10529–10546. <https://doi.org/10.1021/acs.energyfuels.0c02309>

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